Covariant Chronological Kinematics and ZAMO Frame Foliation

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Abstract. We present a covariant kinematic framework for comparing proper clock rates and quantum phase frequencies across stationary, axisymmetric gravitational fields without encountering coordinate-time singularities inside rotating ergospheres. Utilizing the Arnowitt–Deser–Misner (ADM) 3+1 spacetime foliation and Zero-Angular-Momentum Observer (ZAMO) hypersurface normals, we derive three dimensionless chronological invariants: the **ZAMO Dilation Factor (ρ_τ)**, the **Clock Rate Delta ($\Delta\rho_\tau$)**, and the **ZAMO Back-Reaction Factor (H_Q)**. Together with topological wave-crest conservation ($f_{obs} = f_{local} / \rho_\tau$), these quantities replace unphysical Boyer–Lindquist coordinate intervals with strictly causal, observer-relative proper-time ratios valid down to the stretched event horizon.

Keywords: ADM 3+1 Foliation, Zero-Angular-Momentum Observers (ZAMO), Kerr–Newman Spacetime, Proper-Time Dilation, Wave-Crest Conservation, Relativistic Kinematics

AI Collaboration & Authorship Transparency Statement: *This theoretical framework, its mathematical derivations, and its manuscript preparation were developed through an interactive human–AI research collaboration between Luís Pimenta Damásio and Google's Gemini AI assistant. Conceptual direction, physical synthesis, metric architecture, and final editorial verification were directed by the human author, with formal mathematical derivations, LaTeX typesetting, and literature verification generated with the aid of Gemini.*

1. First-Principles Mathematical Derivations

1.1 Covariant Spacetime Metric in ADM 3+1 and Boyer–Lindquist Form

To analyze clock rates and quantum frequency shifts inside rotating ergospheres without coordinate singularities, we express a general stationary, axisymmetric spacetime in Arnowitt–Deser–Misner (ADM) 3+1 form [1, 3]:

$$ds^2 = -N^2 c^2 dt^2 + g_{rr} dr^2 + g_{\theta\theta} d\theta^2 + g_{\varphi\varphi} (d\varphi - \omega_{FD} dt)^2$$

- **ZAMO Lapse Function (N):** In Kerr–Newman geometry of mass M, dimensionless spin a_*, and charge Q_*, $N(\theta) = \sqrt{(\Sigma\Delta / A_{BL})}$ measures the ratio of proper time to asymptotic coordinate time for an observer with zero angular momentum ($L_z = 0$) [2].
- **Frame-Dragging Angular Velocity (ω_{FD}):** Given by $\omega_{FD} = -g_{t\varphi} / g_{\varphi\varphi}$, representing the angular velocity at which spacetime drags Zero-Angular-Momentum Observers around the symmetry axis [2, 3].
- **Unit Timelike ZAMO Normal (n^μ):** Orthogonal to constant-time spatial slices Σ_t, satisfying $g_{\mu\nu} n^\mu n^\nu = -1$, given by $n^\mu = (1/N, 0, 0, \omega_{FD}/N)$.

1.2 Worldline Proper Time and Local Lorentz Factor (γ)

Consider a physical system moving relative to the local ZAMO frame with spatial 3-velocity v (β = v/c) and four-velocity $u^\mu = dx^\mu / d\tau$ obeying $g_{\mu\nu}\, u^\mu\, u^\nu = -c^2$. Projecting u^μ onto n^μ yields the local Lorentz boost factor and proper-time differential:

$$\gamma = -(1/c)\, g_{\mu\nu}\, u^\mu\, n^\nu = 1 / \sqrt{(1 - \beta^2)}, \qquad d\tau_{pre} = [\, \sqrt{N^2}_{pre}(\theta) / \gamma_{pre}\,]\, dt$$

1.3 ZAMO Dilation Factor (ρ_τ) and Clock Rate Delta ($\Delta\rho_\tau$)

Introduce a second physical worldline—the reference observer—located at dimensionless compactness C_{obs} (with static lapse $N^2_{obs} = 1 - C_{obs}$) and moving at speed β_{obs} (Lorentz factor $\gamma_{obs} = (1 - \beta_{obs}{}^2)^{-1/2}$), satisfying $d\tau_{obs} = [\, \sqrt{(1 - C_{obs})} / \gamma_{obs}\,]\, dt$. Eliminating coordinate time dt defines the **ZAMO Dilation Factor (ρ_τ)** and **Clock Rate Delta ($\Delta\rho_\tau$)**:

$$\rho_\tau \equiv d\tau_{obs} / d\tau_{pre} = [\, \gamma_{pre} / \sqrt{N^2}_{pre}(\theta)\,] / [\, \gamma_{obs} / \sqrt{(1 - C_{obs})}\,], \qquad \Delta\rho_\tau \equiv \rho_\tau - 1$$

1.4 Wave-Crest Conservation, Redshift (z), and ZAMO Back-Reaction (H_Q)

For a quantum wavepacket of rest mass m_0 ($f_0 = m_0 c^2 / h$) and proper ZAMO frequency $f_{local} = \gamma_{pre}\, f_0$, topological invariance of elapsed wave cycles $dN = f_{local}\, d\tau_{pre} = f_{obs}\, d\tau_{obs}$ yields:

$$f_{obs} = f_{local} / \rho_\tau, \qquad z \equiv (f_{local} / f_{obs}) - 1 = \Delta\rho_\tau, \qquad H_Q \equiv \sqrt{N^2}_{pre}(\theta) / \gamma_{pre} \in (0, 1]$$

2. Physical Intuition and Operational Interpretation

- **Why Standard Coordinates Fail in Ergospheres:** Static observers ($u^i = 0$) become spacelike inside a rotating black hole's ergosphere ($-g_{tt} \le 0$). Anchoring time ratios to ZAMO worldlines (n^μ) keeps $N^2(\theta) > 0$ everywhere outside the outer horizon r_+, allowing valid clock comparisons down to the stretched-horizon membrane ($N^2 \ge 10^{-6}$) [4].
- **ZAMO Dilation Factor (ρ_τ):** Acts as the exact proper-time exchange rate between two physical clocks. When $\rho_\tau = 1$, both clocks tick synchronously; when $\rho_\tau > 1$, the inspected system experiences gravitational or kinematic slowing; when $\rho_\tau < 1$ (**Inverse Dilation**), the observer sits deeper in a gravity well or moves faster than the target.
- **Clock Rate Delta ($\Delta\rho_\tau$):** Isolates small fractional drifts when $\rho_\tau \approx 1$. In Earth-orbit satellite telemetry, $\Delta\rho_\tau = +4.45 \times 10^{-10}$ directly reveals the +38.5 μs/day net relativistic clock offset.
- **ZAMO Back-Reaction Factor (H_Q):** Merges gravitational time dilation ($\sqrt{N^2}$) and special-relativistic Lorentz suppression ($1/\gamma$) into a single phase-rate gauge between 0 and 1, showing how external observers perceive internal quantum phase evolution freezing as N^2 &to; 10^{-6} or γ &to; ∞.

Acknowledgments

The author gratefully acknowledges Google's Gemini AI assistant for collaborative assistance in formalizing mathematical derivations, verifying dimensional consistency across relativistic limits, and structuring the LaTeX manuscript.

References

1. Arnowitt, R., Deser, S., & Misner, C. W. (1962). The Dynamics of General Relativity. In L. Witten (Ed.), *Gravitation: An Introduction to Current Research* (pp. 227–265). John Wiley & Sons.
2. Bardeen, J. M., Press, W. H., & Teukolsky, S. A. (1972). Rotating Black Holes: Locally Nonrotating Frames, Energy Extraction, and Scalar Synchrotron Radiation. *The Astrophysical Journal*, 178, 347–370. https://doi.org/10.1086/151796
3. Misner, C. W., Thorne, K. S., & Wheeler, J. A. (1973). *Gravitation*. W. H. Freeman and Company.
4. Thorne, K. S., Price, R. H., & MacDonald, D. A. (1986). *Black Holes: The Membrane Paradigm*. Yale University Press.